properties are based on interactions of free electrons with the external sources of energy which may be thermal or magnetic in nature. The application of Maxwell-Boltzmann statistics to this theory allows all the free electrons to gain energy which results in much higher values of heat capacity and paramagnetic susceptibility. As we shall describe later, the use of Fermi-Dirac or quantum statistics allows only a fraction of the total number of free electrons to gain energy and the values of heat capacity and paramagnetic susceptibility thus obtained match with the observed values. These are the principal achievements of the Sommerfeld's theory. The classical theory is also unable to account for the occurrence of long mean free paths (~10^8 to 10^9 interatomic spacings, or more than one centimetre) at low temperatures.

5.2 SOMMERFELD'S QUANTUM THEORY

Considering the free nature of valence electrons as assumed in the classical theory, Sommerfeld treated the problem quantum mechanically using the Fermi-Dirac statistics rather than the classical Maxwell-Boltzmann statistics. The possible electronic energy states in the potential energy box and the distribution of electrons in these states are then determined using quantum statistics. We describe the one-dimensional and three-dimensional cases separately.

5.2.1 Free Electron Gas in One-Dimensional Box

Consider an electron of mass m which is bound to move in a one-dimensional crystal of length L. The electron is prevented from leaving the crystal by the presence of a large potential energy barrier at its surfaces. Although the barriers extend over a few atomic layers near the surface, these are taken infinitely large for the sake of simplicity. The problem is identical to that of an electron moving in a one-dimensional potential box which is represented by a line and is bounded by infinite potential energy barriers as shown in Fig. 5.2. The potential energy within the crystal or box is assumed to be zero. Thus, we have

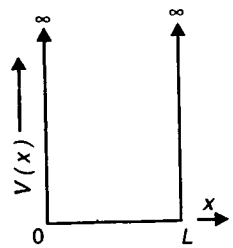


Fig. 5.2 One-dimensional potential box bounded by infinite potential energy barriers.

$$V(x) = 0 \quad \text{for } 0 < x < L$$
$$V(x) = \infty \quad \text{for } x \leq 0 \text{ and } x \geq L \qquad (5.1)$$

The wave function ψ_n of the electron occupying the nth state is determined from the solution of the Schrodinger equation, i.e.,

$$\frac{d^2\psi_n}{dx^2} + \frac{2m}{\hbar^2}(E_n - V)\,\psi_n = 0 \qquad (5.2)$$

where E_n represents the kinetic energy of the electron in the nth state and V is its potential energy. Since $V = 0$ inside the box, Eq. (5.2) becomes

$$\frac{d^2\psi_n}{dx^2} + \frac{2m}{\hbar^2} E_n\psi_n = 0 \tag{5.3}$$

The general solution to this equation is

$$\psi_n(x) = A \sin kx + B \cos kx \tag{5.4}$$

where A and B are arbitrary constants to be determined from the boundary conditions and k is given by

$$k = \sqrt{\frac{2mE_n}{\hbar^2}} \tag{5.5}$$

The boundary conditions are

$$\psi_n(0) = 0 \qquad \text{and} \qquad \psi_n(L) = 0 \tag{5.6}$$

These conditions are based on the fact that at $x = 0$ and L, $V \to \infty$ and the product $V(x)\,\psi_n(x)$ in Eq. (5.2) also approaches infinity. Thus in order that the wave function $\psi_n(x)$ may be continuous, the kinetic energy E_n must also become infinite which is not feasible. Hence $\psi_n(x)$ must vanish for $x = 0$ and L.

For $x = 0$, Eq. (5.4) gives B = 0 and the solution (5.4) becomes

$$\psi_n(x) = A \sin kx \tag{5.7}$$

Also, since $\psi_n(L) = 0$, Eq. (5.7) yields

$$\sin kL = 0$$

or

$$k = \frac{n\pi}{L} \tag{5.8}$$

where n = 1, 2, 3,....Thus the expression (5.7) for the allowed wave function becomes

$$\psi_n(x) = \text{A} \sin\left(\frac{n\pi}{L}\right)x \tag{5.9}$$

The allowed energy values can be obtained from Eqs. (5.5) and (5.8) as

$$E_n = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2 = \frac{n^2h^2}{8mL^2} \tag{5.10}$$

or

$$E_n \propto n^2$$

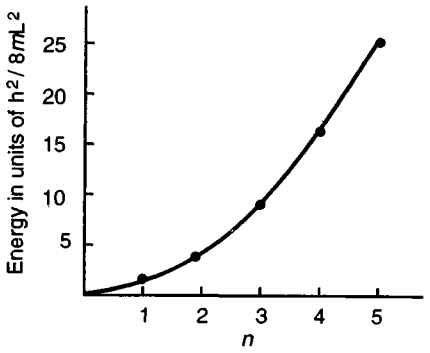


Fig. 5.3. E_n *versus n for a one-dimensional crystal.*

It is apparent from Eqs. (5.9) and (5.10) that the allowed wave functions $\psi_n(x)$ and the allowed energy values E_n exist only for integral values of n. The number n is called the quantum number. Hence the energy spectrum consists of discrete energy levels where the spacing between the levels is determined by the values of n and L. It decreases with increase in L. If L is of the order of a few centimetres, the energy levels form almost a continuum. But if L has atomic dimensions, the spacing between the levels becomes appreciable. The plot of E_n versus n is shown in Fig. 5.3.

The constant A in Eq. (5.9) is determined by using the condition that the probability of finding an electron somewhere on the line is unity, i.e.,

$$\int_0^L \psi_n^*(x)\psi_n(x)dx = 1$$

Using Eq. (5.9), we get

$$A^2 \int_0^L \sin^2\left(\frac{n\pi}{L}x\right)dx = 1$$

or
$$\frac{A^2}{2}\int_0^L\left[1-\cos\left(\frac{2n\pi}{L}x\right)\right]dx = 1$$

or
$$\frac{A^2}{2}\int_0^L dx = 1$$

or
$$A = \sqrt{\frac{2}{L}}$$

Substituting it in Eq. (5.9), we get the normalized wave function as

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi}{L}x\right) \qquad (5.11)$$

The energy levels and the wave functions corresponding to n = 1, 2, 3 and 4 are shown in Fig. 5.4.

We now discuss the distribution of electrons among the allowed energy levels and determine the Fermi energy and the density of states.

(i) Fermi Energy

The electrons are distributed among the various possible energy levels in accordance with the Pauli's exclusion principle which states that no two electrons can have all their quantum numbers identical, i.e., each orbital or quantum state can be occupied by at the most one electron. But the different states may have different values (non-degenerate case) or the same value (degenerate case) of energy. In a solid, an electron in a conduction electronic state has the quantum numbers n and m_s where n is the principal quantum number and m_s is the magnetic spin quantum number. Each set of values of n and m_s define a quantum state. For each value of n, m_s can have two possible values, +1/2 or −1/2. This means that each energy level defined by the quantum number n can have two quantum states and hence can accommodate a maximum of two electrons, one with spin up and the other with spin down. In other words, each energy level is doubly degenerate. For example, if there are seven electrons of appropriate spins then, in the ground state, these can be accommodated in four energy levels; the first three levels would have two electrons each with opposite spins and the fourth level would contain the last unpaired electron. These four energy levels are represented by the values 1, 2, 3 and 4 of the quantum number n. Thus, we find that if the total number of electrons to be accommodated is seven, the energy levels with $n < 4$ would be occupied while the level with $n > 4$ would be empty. The topmost filled energy level at 0 K is known as the *Fermi level* and the energy corresponding to this level is called the *Fermi energy*, E_F.

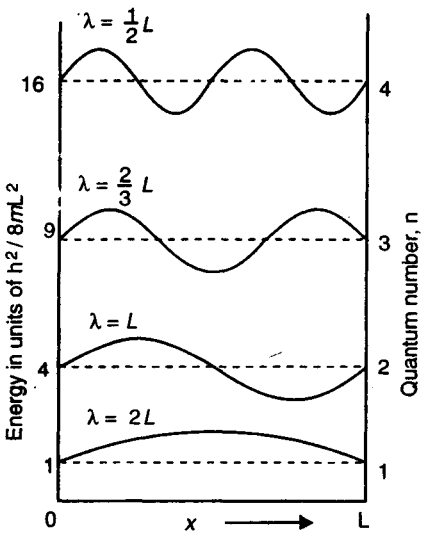


Fig. 5.4. *First four wave functions (solid lines) and the corresponding energy levels (broken lines) of an electron in a one-dimensional crystal.*

If N is the total number of electrons to be accommodated on the line then, for even n, we can write

$$2n_F = N \tag{5.12}$$

where n_F represents the principal quantum number of the Fermi level. Thus, for n = n_F, Eq. (5.10) gives

$$E_F = \frac{\hbar^2}{2m}\left(\frac{n_F\pi}{L}\right)^2 = \frac{\hbar^2}{2m}\left(\frac{N\pi}{2L}\right)^2 \tag{5.13}$$

Thus the value of the Fermi energy depends upon the length of the box the number of electrons in the box. For example, taking

$$\frac{N}{L} = 0.5 \text{ electrons/Å or } 5 \times 10^7 \text{ electrons/cm},$$

the Eq. (5.13) gives

$$E_F = 3.7 \times 10^{-12} \text{ erg} = 2.4 \text{ eV}$$

Thus if we accommodate 5×10^7 electrons on one centimetre length of the line, the energy of the topmost electron would be 2.4 eV.

(ii) Total Energy

The total energy, E_o, of all the N electrons in the ground state is determined by summing up the energies of the individual electrons. For N electrons, the number of filled energy levels is $N/2$ and E_0 is given by

$$E_0 = 2 \sum_{n=1}^{N/2} E_n$$

Here the factor of 2 appears because each level contains two electrons with equal energies. Using Eq. (5.10), we get

$$E_0 = 2\,\frac{\hbar^2}{2m}\left(\frac{\pi}{L}\right)^2 \sum_{n=1}^{N/2} n^2 \tag{5.14}$$

Since

$$\sum_{n=1}^{s} n^2 = \frac{1}{6}s\left(2s^2+3s+1\right)$$

$$\cong \frac{1}{3}s^3 \qquad \text{for} \quad s \geq 1$$

$$\therefore \qquad \sum_{n=1}^{N/2} n^2 \cong \frac{1}{3}\left(\frac{N}{2}\right)^3$$

Hence Eq. (5.14) becomes

$$E_0 = 2\,\frac{\hbar^2}{2m}\left(\frac{\pi}{L}\right)^2 \frac{1}{3}\left(\frac{N}{2}\right)^3 = \frac{1}{3}\,\frac{\hbar^2}{2m}\left(\frac{N\pi}{2L}\right)^2 N$$

or, using Eq. (5.13), we obtain

$$E_0 = \frac{1}{3}NE_F \tag{5.15}$$

Thus, for one-dimensional crystal, the average kinetic energy in the ground state is one-third of the Fermi energy.

(iii) Density of States

The density of states is defined as the number of electronic states present in a unit energy range. It is denoted by $D(E)$ and is given by

$$D(E) = \frac{dn}{dE} \tag{5.16}$$

where dn represents the number of electronic quantum states present in the energy interval E and $E + dE$. For a free electron gas, since each energy level contains two electronic states, one with spin up and the other with spin down, the actual density of states is twice the value given by Eq. (5.16). Thus, we have

$$D(E) = 2\frac{dn}{dE} \tag{5.17}$$

From. Eq. (5.10), we obtain

$$\frac{dE}{dn} = \frac{\hbar^2}{2m}\left(\frac{\pi}{L}\right)^2 2n = \frac{h^2 n}{4mL^2}$$

Therefore, Eq. (5.17) becomes

$$D(E) = 2\left(\frac{4mL^2}{nh^2}\right) = \frac{8mL^2}{h^2}\,\frac{1}{n}$$

Again, from Eq. (5.10), we get

$$\frac{1}{n} = \left(\frac{h^2}{8mL^2 E}\right)^{1/2}$$

Hence

$$D(E) = \left(\frac{8mL^2}{h^2 E}\right)^{1/2} = \frac{4L}{h}\left(\frac{m}{2E}\right)^{1/2} \tag{5.18}$$